

**EGERTON UNIVERSITY
FACULTY OF SCIENCE
DEPARTMENT OF COMPUTER SCIENCE**

**RESEARCH PROPOSAL
ANALYSIS OF PHYSICAL NOISE EFFECTS ON CORRECTNESS AND EFFICIENCY OF
QUANTUM ALGORITHMS**

**Submitted by:
KIDENGE ELISHA
Registration Number: SM24/11854/24**

**Supervisors:
DR. NICASIO MAGUU
Lecturer,
Department of Electrical and Control Engineering
Egerton University
DR. DUKE OEBA
Lecturer & Researcher, Renewable Energy
Department of Physics
Egerton University**

**Proposal submitted in partial fulfillment
of the requirements for the degree of
MASTER OF SCIENCE
In
COMPUTER SCIENCE
March 2026**

DECLARATION AND APPROVAL

I, Elisha Odhiambo Kidenge, hereby declare that this research proposal:

1. Represents my original work
2. Has not been submitted for any other degree at Egerton University or elsewhere
3. Contains all references and acknowledgments of sources used

Student:

Elisha Odhiambo Kidenge
Registration No: SM24/11854/24
Date:

Supervisors' Approval:

Dr. Nicasio Maguu Muchuka
Department of Electrical and Control Engineering
Egerton University
Date:

Dr. Duke Oeba
Department of Physics
Egerton University
Date:

Chairman, Department of Computer Science:

Date:

TABLE OF CONTENTS

DECLARATION AND APPROVAL	ii
TABLE OF CONTENTS	iii
LIST OF FIGURES	iv
1. TOPIC OVERVIEW	1
2. KEY CONCEPTS	1
2.1 QUANTUM NOISE	1
2.2 CORRECTNESS & EFFICIENCY	1
3. OBJECTIVES (ALIGNED TO BLOOM’S TAXONOMY)	2
4. RESEARCH QUESTIONS	2
5. METHODOLOGY	2
5.1 Step 1: Noise Modeling	2
5.2 Step 2: Algorithm Simulation	2
5.3 Step 3: Performance Evaluation	2
5.4 Step 4: Error Analysis & Mitigation	2
6. KEY TOOLS	3
7. EXPECTED OUTCOMES.....	3
8. BIBLIOGRAPHY	3

LIST OF FIGURES

LIST OF TABLES

Here's a structured set of **concise notes** for my MSc topic: "*Analysis of Physical Noise Effects on Correctness and Efficiency of Quantum Algorithms*".

1. TOPIC OVERVIEW

- **Field:** Quantum Computing & Quantum Information[1], [2], [3], [4], [5], [6], [7], [8], [9], [10], [11], [12], [13], [14], [15], [16], [17], [18], [19], [20], [21], [22], [23], [24], [25], [26], [27]
- **Focus:** Investigating how **physical noise** affects the **performance and correctness** of quantum algorithms.
- **Algorithms of Interest:**
 - **Grover's Algorithm** – used for unstructured search problems.
 - **Quantum Fourier Transform (QFT)** – used in factoring, period finding, and quantum phase estimation.
- **Simulation Platform: Qiskit** – IBM's open-source quantum computing framework.

2. KEY CONCEPTS

2.1 QUANTUM NOISE

- Quantum systems are sensitive to **environmental interactions**, leading to errors.
- Types of physical noise:
 - **Bit-flip error** – qubit flips from $|0\rangle \leftrightarrow |1\rangle$.
 - **Phase-flip error** – relative phase of qubit changes.
 - **Depolarizing error** – qubit loses coherence, becomes mixed state.
 - **Amplitude damping** – qubit relaxes to ground state over time.
- Noise is modelled using **Kraus operators** or **quantum channels**.

2.2 CORRECTNESS & EFFICIENCY

- **Correctness:** Probability that algorithm outputs the correct solution.
- **Efficiency:** Number of qubits, gates, or computational steps required; impact of noise can increase runtime or reduce accuracy.

3. OBJECTIVES (ALIGNED TO BLOOM'S TAXONOMY)

1. **Remember & Understand:** Identify and classify common physical noise sources in quantum systems.
2. **Apply:** Implement Grover's Algorithm and QFT in Qiskit under simulated noise conditions.
3. **Analyze:** Evaluate how different noise types affect algorithm correctness and efficiency.
4. **Evaluate:** Compare performance degradation across algorithms and noise models.
5. **Create:** Propose mitigation strategies or error models to improve algorithm reliability.

4. RESEARCH QUESTIONS

1. What are the dominant physical noise sources in practical quantum systems?
2. How does each type of noise affect the correctness of Grover's Algorithm and QFT?
3. How does noise impact the computational efficiency of these algorithms?
4. Can simulation-based error modeling help in designing more robust quantum algorithms?

5. METHODOLOGY

5.1 Step 1: Noise Modeling

- ✓ Use **Kraus operator formalism** to represent quantum noise channels.
- ✓ Simulate **bit-flip, phase-flip, depolarizing, and amplitude damping** errors.

5.2 Step 2: Algorithm Simulation

- ✓ Implement Grover's Algorithm and QFT in **Qiskit**.
- ✓ Introduce noise channels at different stages of computation.

5.3 Step 3: Performance Evaluation

- ✓ Measure **correctness**: probability of correct output.
- ✓ Measure **efficiency**: gate count, circuit depth, runtime.
- ✓ Analyze trends across different noise levels.

5.4 Step 4: Error Analysis & Mitigation

- ✓ Identify which noise types are most detrimental.
- ✓ Suggest or simulate **basic error correction techniques** or **algorithm adjustments**.

6. KEY TOOLS

- **Qiskit** (IBM Quantum)
- Python for simulations
- Matplotlib/Seaborn for plotting algorithm performance
- Statistical metrics: fidelity, error rate, success probability

7. EXPECTED OUTCOMES

- Quantitative assessment of **how noise affects algorithm correctness and efficiency**.
- Identification of **noise-resilient strategies** for Grover and QFT.
- Insights for **future quantum hardware optimization** and algorithm design.

8. BIBLIOGRAPHY

- [1] IBM Quantum, “IBM Quantum Processor Documentation.” 2023.
- [2] L. K. Grover, “A fast quantum mechanical algorithm for database search,” in *Proceedings of the 28th Annual ACM Symposium on Theory of Computing*, 1996, pp. 212–219. doi: 10.1145/237814.237866.
- [3] E. Farhi, J. Goldstone, and S. Gutmann, “A quantum approximate optimization algorithm,” *arXiv preprint*, 2014.
- [4] D. Arias, O. Mentxaka, E. Terres, and P. García Bringas, “Assessing the Impact of Noise on Quantum Neural Networks: An Experimental Analysis,” *arXiv preprint*, 2023.
- [5] J. R. McClean, S. Boixo, V. N. Smelyanskiy, R. Babbush, and H. Neven, “Barren plateaus in quantum neural network training landscapes,” *Nature Communications*, vol. 9, no. 1, p. 4812, 2018, doi: 10.1038/s41467-018-07090-4.
- [6] J. M. Gambetta, J. M. Chow, and M. Steffen, “Building logical qubits in a superconducting quantum computing system,” *npj Quantum Information*, vol. 3, no. 1, pp. 1–7, 2017, doi: 10.1038/s41534-016-0004-0.
- [7] R. Blume-Kohout, J. K. Gamble, E. Nielsen, J. Mizrahi, J. D. Sterk, and P. Maunz, “Demonstration of qubit operations below a rigorous fault tolerance threshold with gate set tomography,” *Nature Communications*, vol. 8, no. 1, p. 14485, 2017, doi: 10.1038/ncomms14485.
- [8] D. C. McKay, C. J. Wood, S. Sheldon, J. M. Chow, and J. M. Gambetta, “Efficient Z gates for quantum computing,” *Physical Review A*, vol. 97, no. 2, p. 022332, 2018, doi: 10.1103/PhysRevA.97.022332.

- [9] Y. Sung, Z. Ding, O. Moussa, J. M. Chow, and J. M. Gambetta, “Exploiting noise as a resource for computation and learning in quantum systems,” *Physical Review X*, vol. 10, no. 1, p. 011017, 2020, doi: 10.1103/PhysRevX.10.011017.
- [10] A. Kandala *et al.*, “Hardware-efficient variational quantum eigensolver for small molecules and quantum magnets,” *Nature*, vol. 549, no. 7671, pp. 242–246, 2017, doi: 10.1038/nature23879.
- [11] S. Endo, Z. Cai, S. C. Benjamin, and X. Yuan, “Hybrid quantum-classical algorithms and quantum error mitigation,” *Journal of the Physical Society of Japan*, vol. 90, no. 3, p. 032001, 2021, doi: 10.7566/JPSJ.90.032001.
- [12] L. Cincio, Y. Subaşı, A. T. Sornborger, and P. J. Coles, “Learning the quantum algorithm for state overlap,” *New Journal of Physics*, vol. 20, no. 11, p. 113022, 2018, doi: 10.1088/1367-2630/aae94a.
- [13] P. Magnard *et al.*, “Measurement and control of a superconducting quantum processor with a fully integrated radio-frequency system on a chip,” *Physical Review Applied*, vol. 14, no. 5, p. 054026, 2020, doi: 10.1103/PhysRevApplied.14.054026.
- [14] T. Kumar, D. Kumar, and G. Singh, “Noise Analysis of Grover’s Quantum Search Algorithm,” *Indian Journal of Pure & Applied Physics*, vol. 61, pp. 355–365, 2023, doi: 10.56042/ijpap.v61i5.70285.
- [15] Forum ICMUV, “Noise in Quantum Computing: Spin-Boson Model and Lindblad Master Equation.” 2023.
- [16] K. Sharma, S. Khatri, M. Cerezo, and P. J. Coles, “Noise resilience of variational quantum compiling,” *New Journal of Physics*, vol. 22, no. 4, p. 043006, 2020, doi: 10.1088/1367-2630/ab784c.
- [17] S. Wang *et al.*, “Noise-induced barren plateaus in variational quantum algorithms,” *arXiv preprint*, 2020.
- [18] J. Kelly *et al.*, “Optimal quantum control using randomized benchmarking,” *Physical Review Letters*, vol. 112, no. 24, p. 240504, 2014, doi: 10.1103/PhysRevLett.112.240504.
- [19] Qiskit Development Team, “Qiskit: An Open-source Framework for Quantum Computing.” 2023.
- [20] M. A. Nielsen and I. L. Chuang, *Quantum Computation and Quantum Information: 10th Anniversary Edition*. Cambridge: Cambridge University Press, 2010.
- [21] J. Preskill, “Quantum Computing in the NISQ era and beyond,” *Quantum*, vol. 2, p. 79, 2018, doi: 10.22331/q-2018-08-06-79.

- [22] E. Knill, “Quantum computing with realistically noisy devices,” *Nature*, vol. 434, no. 7029, pp. 39–44, 2005, doi: 10.1038/nature03350.
- [23] Z. Cai *et al.*, “Quantum error mitigation,” *arXiv preprint*, 2020.
- [24] D. Volya and P. Mishra, “Special Session: Impact of Noise on Quantum Algorithms in Noisy Intermediate-Scale Quantum Systems,” in *IEEE/ACM International Conference on Computer-Aided Design (ICCAD)*, 2020. doi: 10.1145/3400302.3415753.
- [25] M. H. Devoret and R. J. Schoelkopf, “Superconducting circuits for quantum information: An outlook,” *Science*, vol. 339, no. 6124, pp. 1169–1174, 2013, doi: 10.1126/science.1231930.
- [26] R. Barends *et al.*, “Superconducting quantum circuits at the surface code threshold for fault tolerance,” *Nature*, vol. 508, no. 7497, pp. 500–503, 2014, doi: 10.1038/nature13171.
- [27] A. W. Cross, L. S. Bishop, S. Sheldon, P. D. Nation, and J. M. Gambetta, “Validating quantum computers using randomized model circuits,” *arXiv preprint*, 2018.